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Analysis of the Influencing Factors of Road Environment in Road Traffic Accidents

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***Abstract*—Road traffic is an important mode of transportation for daily travel. Traffic accidents cause tens of thousands of deaths every year. Studying the influencing factors in traffic accidents will help to discover the main causes of accidents and prevent the recurrence of similar accidents. Among the factors that caused the accident, the road environment factors played an important role. The paper focuses on the impact of road environmental factors on road traffic accidents and analyzes the research results of historical documents. Based on road traffic accident statistical annual report data, the study of the impact of road environmental factors on accidents in road traffic accidents provides a basis for ensuring driving safety and analyzing the factors affecting accidents.**

***Keywords*-Traffic accidents; Road transportation; Causes of accidents; Road environment; Accident influence factors**

I. INTRODUCTION

As an important means of transportation, road transportation plays an important role in people's transportation and daily life. With the increasing frequency of business activities, tourism and cargo transportation activities, the safety of road traffic has become an important issue for every transportation activity. According to statistics, there are hundreds of thousands of traffic accidents every year, with tens of thousands of deaths, causing huge losses and serious injuries to people and society.

Road transportation includes multiple intermediate links such as preparation, loading and unloading, and transportation. Road traffic accidents mostly occur during road transportation. The safety of road transportation is related to many factors, including driver factors, vehicle body factors, road environment factors, and business management factors. Based on the research results of historical literature and statistical annual report data, this paper analyzes the impact of road environment on accidents in road traffic accidents, and provides a basis for the construction of safety evaluation indicators for road transportation.

II. ROAD ENVIRONMENTAL FACTORS

Domestic researchers have conducted a lot of research on safety factors that affect road transportation, including road environmental factors. Using on-site investigation, questionnaire survey, and association rules to analyze the factors affecting the accident, it is found that the main causes of road traffic accidents are human factors, mechanical failures and environmental factors, and the most influential factors are human factors and road types [1-2]. The factors affecting urban road traffic safety are analyzed from the aspects of human factors, vehicle factors, road factors, environmental factors and management factors [3-4]. Using statistical analysis method, road contour buffer analysis method and other methods, the law and characteristics of accidents are analyzed, and the key factors affecting traffic accidents are determined [5-12]. Using data mining methods to study potential laws and accident mechanisms, the research shows that "rainy days" and "night (no lighting)" are the most likely factors to cause serious injury accidents [13]. The influencing factors of urban ice and snow road traffic accidents were screened, and the causes of road traffic accidents were determined [14].

The statistical data of road traffic accidents can reflect the degree of aggregation of various road traffic accidents. Accident types with a large number of accidents and fatalities may cause greater harm to drivers and passengers. The following is an analysis of the impact of road environmental factors on the annual report statistics to discover the degree of impact of various factors on road traffic accidents.

A. Road Type

Table 1 shows the proportion of accidents, deaths, and injuries by road types. It can be seen that there is not much difference between the number of accidents and the number of injured under highway and city road conditions, but the number of deaths on highways is more than twice the number of deaths on city roads due to the fast speed of vehicles. Therefore, the hazard degree of accidents on highways is much higher than that of city roads.

TABLE I. THE PROPORTION OF ACCIDENTS, DEATHS, AND INJURIES BY ROAD TYPES

Items	Proportion of accidents	Percentage of deaths	Proportion of injured
Highway	52.98%	67.01%	54.80%
City road	47.02%	32.99%	45.20%

Fig. 1 shows the proportion of accidents, deaths, and injuries of various road types in highways. It can be seen that Level-2 roads account for the largest proportion, and the statistical data of Level-3 roads, Level-4 roads and Level-5 roads have little difference. Although there are not many accidents on expressways, the number of fatalities in a single accident far exceeds that of other road types. Therefore, vehicles driving on expressways should be strictly monitored to avoid major traffic accidents.

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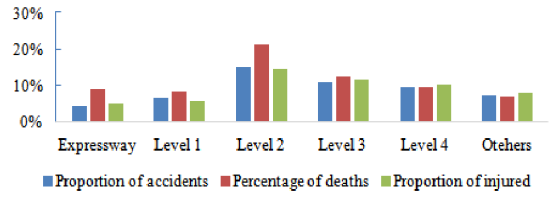

Figure 1. (Highway) The proportion of accidents, deaths, and injuries

Fig. 2 shows the proportion of accidents, deaths, and injuries of various road types in city roads. It can be seen that general urban roads account for the largest proportion, followed by other types and urban expressways.

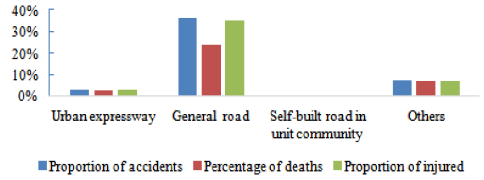

Figure 2. (City roads) The proportion of accidents, deaths, and injuries by road type

B. Highway Administrative Level

Fig. 3 shows the proportion of accidents, fatalities, and injuries by highway administrative level. It can be seen that the statistical data of each highway administrative level is not much different, indicating that the impact of the highway administrative level on the accident is negligible.

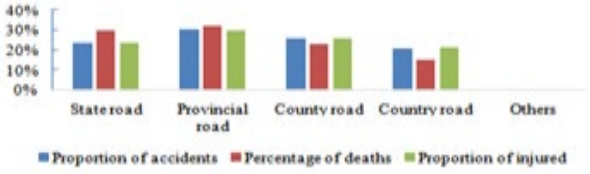

Figure 3. The proportion of accidents, deaths, and injuries by highway administrative level

C. Terrain Conditions

Table 2 shows the proportion of accidents, deaths, and injuries by terrain conditions. The number of accidents on flat roads is the largest, which is related to the high driving speed and the low safety awareness of drivers. There is little difference in the number of deaths and the number of injured in a single accident.

TABLE II. THE PROPORTION OF ACCIDENTS, DEATHS, AND INJURIES BY TERRAIN CONDITIONS

Items	Proportion of accidents	Percentage of deaths	Proportion of injured
Plain	66.40%	61.75%	63.62%
Hills	16.46%	17.50%	16.94%
Mountains	17.14%	20.76%	19.44%

D. Road Alignment

Table 3 shows the proportion of accidents, deaths, and injuries based on road alignment. Straight sections on flat roads are a high-risk area for serious accidents due to high vehicle speeds. Curved road sections and sloping road sections are also areas with a high incidence of accidents. Therefore, the influence of road alignment factors on driving safety should be considered.

TABLE III. THE PROPORTION OF ACCIDENTS, DEATHS, AND INJURIES BY ROAD ALIGNMENT

Items	Proportion of accidents	Percentage of deaths	Proportion of injured
Straight	79.84%	73.97%	78.20%
General bend	10.85%	13.48%	11.46%
General slope	2.67%	2.83%	2.96%
Sharp bend	0.47%	0.61%	0.59%
Steep slope	0.18%	0.30%	0.20%
Slopes	0.17%	0.30%	0.20%
General curved slope	4.36%	6.47%	4.72%
Sharp bends and steep slopes	0.67%	1.06%	0.77%
General slopes and sharp bends	0.46%	0.51%	0.56%
General bends and steep slopes	0.32%	0.47%	0.34%

E. Road Traffic Conditions

Fig. 4 shows the proportion of accidents, deaths, and injuries by road traffic conditions. It can be seen that 85% of accidents occur under dry road conditions, which is related to factors such as driver speeding and irregular operation. Under poor road traffic conditions, the number of accidents on wet roads accounted for the largest proportion, followed by water and snow road conditions. The impact of wet and slippery road conditions on driving safety should be considered.

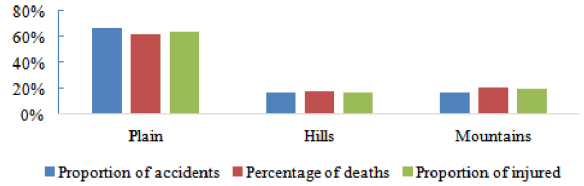

Figure 4. The proportion of accidents, deaths, and injuries by road traffic conditions

F. Lighting Conditions

Fig. 5 shows the proportion of accidents, deaths, and injuries by lighting conditions. The number of fatalities in a single accident during the day is the lowest, followed by lighting conditions at night. The death toll for a single accident is highest at night, followed by dawn and dusty weather. Therefore, the impact of lights on driving safety should be considered.

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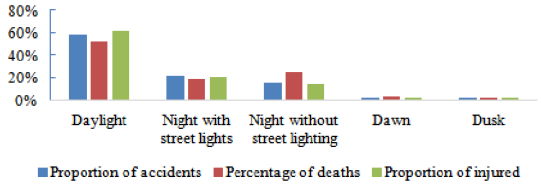

Figure 5. The proportion of accidents, deaths, and injuries calculated by lighting conditions

G. Visibility Conditions

Fig. 6 shows the proportion of accidents, deaths, and injuries by visibility conditions. It can be seen that the visibility of more than 200 meters to the number of accidents accounted for the largest proportion. The cause of the accident may be related to the excessive speed or the driver's failure to implement safe driving operation specifications. The number of fatalities in a single accident within 50 meters of visibility is the highest, and risk management and control of vehicles should be strengthened when the visibility is very low.

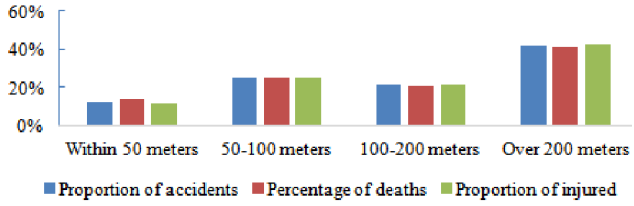

Figure 6. The proportion of accidents, deaths, and injuries based on visibility conditions

H. Weather Conditions

Fig. 7 shows the proportion of the number of accidents, deaths, and injuries by weather conditions. Among the factors in bad weather conditions, cloudy and rainy days accounted for the largest proportion of accidents, followed by snow and fog weather. Weather conditions such as rain, snow, and fog will adversely affect the driver's line of sight and road conditions, and affect the safe driving of vehicles.

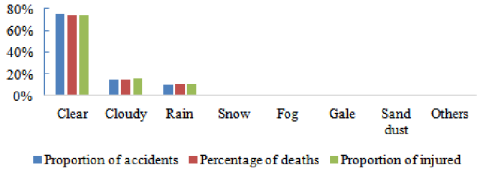

Figure 7. The proportion of accidents, deaths, and injuries by weather conditions

III. CONCLUSION

Analyzing the historical research results and the data in the annual road traffic accident statistics report, it can be seen that road environmental factors are one of the evaluation indicators that affect the safety of road transport vehicles. The impact of road environmental factors such as road type, road alignment, road traffic conditions, lighting conditions, visibility conditions and weather conditions on driving safety should be considered in the road transportation safety evaluation. When the visibility is less than 50 meters, roads with no lights at night or at night, or slippery roads, the risk of driving is significantly increased.

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